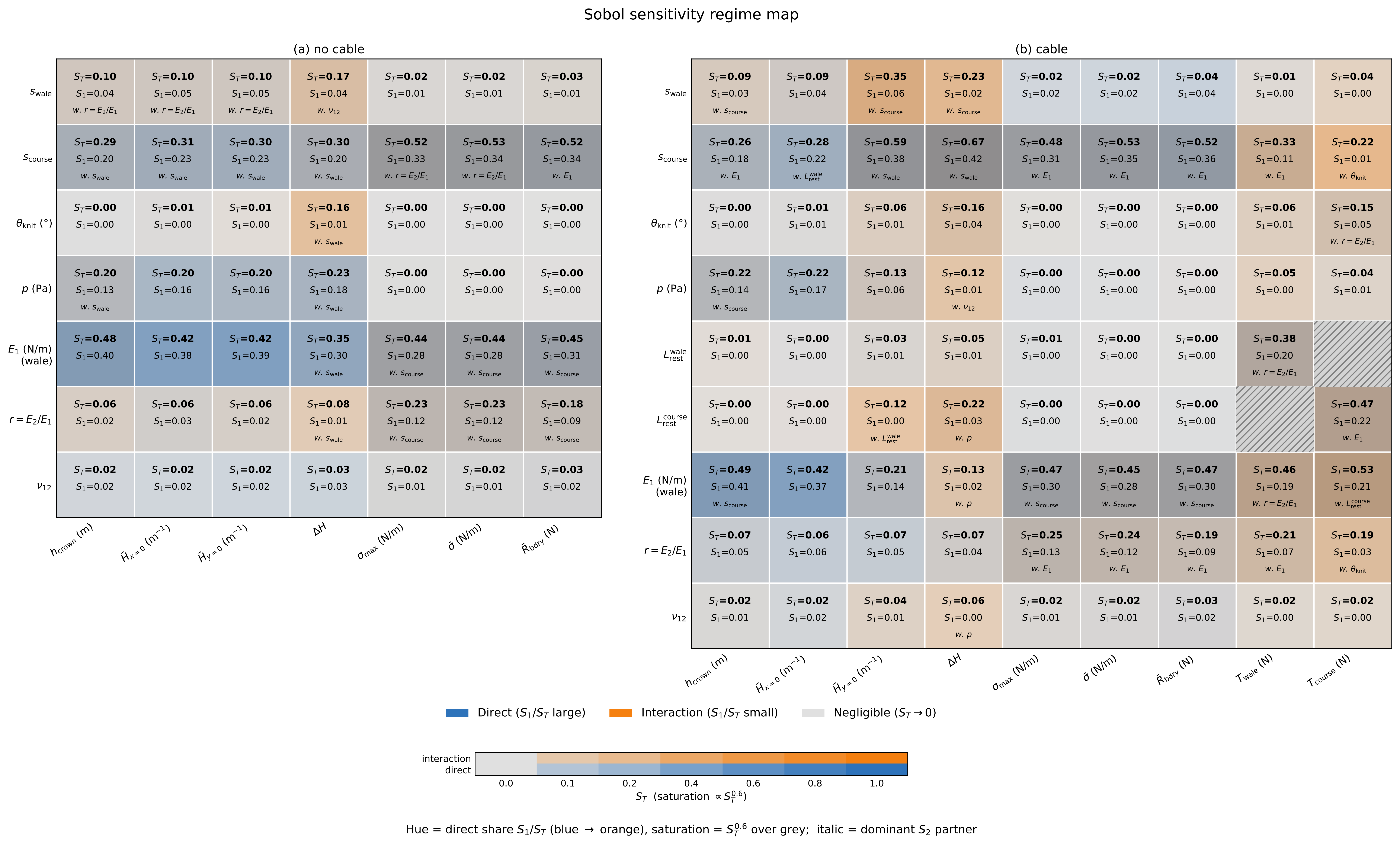
Sobol sensitivity regime map
(a) no cable
s_{wale}: S_T=0.10, S_1=0.04, w. $r=E_2/E_1$ | S_T=0.10, S_1=0.05, w. $r=E_2/E_1$ | S_T=0.10, S_1=0.05, w. $r=E_2/E_1$ | S_T=0.17, S_1=0.04, w. ν_{12} | S_T=0.02, S_1=0.01 | S_T=0.02, S_1=0.01 | S_T=0.03, S_1=0.01
s_{course}: S_T=0.29, S_1=0.20, w. s_{wale} | S_T=0.31, S_1=0.23, w. s_{wale} | S_T=0.30, S_1=0.23, w. s_{wale} | S_T=0.30, S_1=0.20, w. s_{wale} | S_T=0.52, S_1=0.33, w. $r=E_2/E_1$ | S_T=0.53, S_1=0.34, w. $r=E_2/E_1$ | S_T=0.52, S_1=0.34, w. E_1
θ_{knit} (°): S_T=0.00, S_1=0.00 | S_T=0.01, S_1=0.00 | S_T=0.01, S_1=0.00 | S_T=0.16, S_1=0.01, w. s_{wale} | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00
p (Pa): S_T=0.20, S_1=0.13, w. s_{wale} | S_T=0.20, S_1=0.16 | S_T=0.20, S_1=0.16 | S_T=0.23, S_1=0.18, w. s_{wale} | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00
E_1 (N/m) (wale): S_T=0.48, S_1=0.40 | S_T=0.42, S_1=0.38 | S_T=0.42, S_1=0.39 | S_T=0.35, S_1=0.30, w. s_{wale} | S_T=0.44, S_1=0.28, w. s_{course} | S_T=0.44, S_1=0.28, w. s_{course} | S_T=0.45, S_1=0.31, w. s_{course}
$r=E_2/E_1$: S_T=0.06, S_1=0.02 | S_T=0.06, S_1=0.03 | S_T=0.06, S_1=0.02 | S_T=0.08, S_1=0.01, w. s_{wale} | S_T=0.23, S_1=0.12, w. s_{course} | S_T=0.23, S_1=0.12, w. s_{course} | S_T=0.18, S_1=0.09, w. s_{course}
ν_{12}: S_T=0.02, S_1=0.02 | S_T=0.02, S_1=0.02 | S_T=0.02, S_1=0.02 | S_T=0.03, S_1=0.03 | S_T=0.02, S_1=0.01 | S_T=0.02, S_1=0.01 | S_T=0.03, S_1=0.02
h_{crown} (m) | $\bar{H}_{x=0}$ (m^{-1}) | $\bar{H}_{y=0}$ (m^{-1}) | ΔH | $\sigma_{\max}$ (N/m) | $\bar{\sigma}$ (N/m) | $\bar{R}_{\text{bdry}}$ (N)
(b) cable
s_{wale}: S_T=0.09, S_1=0.03, w. s_{course} | S_T=0.09, S_1=0.04 | S_T=0.35, S_1=0.06, w. s_{course} | S_T=0.23, S_1=0.02, w. s_{course} | S_T=0.02, S_1=0.02 | S_T=0.02, S_1=0.02 | S_T=0.04, S_1=0.04 | S_T=0.01, S_1=0.00 | S_T=0.04, S_1=0.00
s_{course}: S_T=0.26, S_1=0.18, w. E_1 | S_T=0.28, S_1=0.22, w. $L_{\text{rest}}^{\text{wale}}$ | S_T=0.59, S_1=0.38, w. s_{wale} | S_T=0.67, S_1=0.42, w. s_{wale} | S_T=0.48, S_1=0.31, w. E_1 | S_T=0.53, S_1=0.35, w. E_1 | S_T=0.52, S_1=0.36, w. E_1 | S_T=0.33, S_1=0.11, w. E_1 | S_T=0.22, S_1=0.01, w. θ_{knit}
θ_{knit} (°): S_T=0.00, S_1=0.00 | S_T=0.01, S_1=0.01 | S_T=0.06, S_1=0.01 | S_T=0.16, S_1=0.04 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.06, S_1=0.01 | S_T=0.15, S_1=0.05, w. $r=E_2/E_1$
p (Pa): S_T=0.22, S_1=0.14, w. s_{course} | S_T=0.22, S_1=0.17 | S_T=0.13, S_1=0.06 | S_T=0.12, S_1=0.01, w. ν_{12} | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.05, S_1=0.00 | S_T=0.04, S_1=0.01
$L_{\text{rest}}^{\text{wale}}$: S_T=0.01, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.03, S_1=0.01 | S_T=0.05, S_1=0.01 | S_T=0.01, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.38, S_1=0.20, w. $r=E_2/E_1$ |
$L_{\text{rest}}^{\text{course}}$: S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.12, S_1=0.00, w. $L_{\text{rest}}^{\text{wale}}$ | S_T=0.22, S_1=0.03, w. p | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | S_T=0.00, S_1=0.00 | | S_T=0.47, S_1=0.22, w. E_1
E_1 (N/m) (wale): S_T=0.49, S_1=0.41, w. s_{course} | S_T=0.42, S_1=0.37 | S_T=0.21, S_1=0.14 | S_T=0.13, S_1=0.02, w. p | S_T=0.47, S_1=0.30, w. s_{course} | S_T=0.45, S_1=0.28, w. s_{course} | S_T=0.47, S_1=0.30, w. s_{course} | S_T=0.46, S_1=0.19, w. $r=E_2/E_1$ | S_T=0.53, S_1=0.21, w. $L_{\text{rest}}^{\text{course}}$
$r=E_2/E_1$: S_T=0.07, S_1=0.05 | S_T=0.06, S_1=0.06 | S_T=0.07, S_1=0.05 | S_T=0.07, S_1=0.04 | S_T=0.25, S_1=0.13, w. E_1 | S_T=0.24, S_1=0.12, w. E_1 | S_T=0.19, S_1=0.09, w. E_1 | S_T=0.21, S_1=0.07, w. E_1 | S_T=0.19, S_1=0.03, w. θ_{knit}
ν_{12}: S_T=0.02, S_1=0.01 | S_T=0.02, S_1=0.02 | S_T=0.04, S_1=0.01 | S_T=0.06, S_1=0.00, w. p | S_T=0.02, S_1=0.01 | S_T=0.02, S_1=0.01 | S_T=0.03, S_1=0.02 | S_T=0.02, S_1=0.00 | S_T=0.02, S_1=0.00
h_{crown} (m) | $\bar{H}_{x=0}$ (m^{-1}) | $\bar{H}_{y=0}$ (m^{-1}) | ΔH | $\sigma_{\max}$ (N/m) | $\bar{\sigma}$ (N/m) | $\bar{R}_{\text{bdry}}$ (N) | T_{wale} (N) | T_{course} (N)
Direct (S_1/S_T large)
Interaction (S_1/S_T small)
Negligible ($S_T \to 0$)
interaction
direct
0.0 0.1 0.2 0.4 0.6 0.8 1.0
S_T (saturation $\propto S_T^{0.6}$)
Hue = direct share S_1/S_T (blue → orange), saturation = $S_T^{0.6}$ over grey; italic = dominant S_2 partner